# Restored LyphoSAT SAT constructors: terminal regression result

Tactix development checkpoint

October 4, 2026

Frozen checkpoint 156afef6b is not accepted. B28+C5 passes, but Rank1--50 exceeds its aggregate solver-time ceiling and WV100/WV2 lose seven retained SAT solves. All 272 rows are terminal. There is no wrong verdict, rejected model, rejected proof, or missing result.

| Set | Gate | Solver seconds | Checked | Censored | Timeouts | Lost |
|---|---|---|---|---|---|---|
| B28+C5 | pass | 998.177 / 999.876 | 32 | 1 | 0 | 0 |
| Rank1--50 | fail | 9,530.106 / 8,521.522 | 42 | 8 | 0 | 0 |
| WV100 | fail | 2,646.841 | 93 | 2 | 5 | 5 |
| WV2--89 | fail | 3,781.591 | 75 | 2 | 12 | 2 |
| Total | fail | | 242 | 13 | 17 | 7 |

Solver limits are 1,200 seconds except WV2's 600 seconds. Proof checking is bounded at 1,800 seconds. Twelve censored checks satisfy the agreed cost bound; the WV100 ncc row is inconclusive because its baseline was already censored. Censoring is not independent proof verification, but it is not checker failure.

## Diagnosis

The frozen DefaultPolicy omitted historical affine constructor instance 8. All seven lost retained solves and the three largest Rank regressions have one baseline affine call and no candidate affine call. Six MDP rows were solved directly by the old affine constructor. Rank cabp-V-nos6, WV100 Break_20_72, and the WV2 timetable entered affine without producing a model, so their association is trajectory evidence rather than proof of a direct constructor solution.

Mutable commit 60446aa0d restores instance 8 after exact-one admission and before structure-changing preprocessing. Its policy-path test requires the productive case to solve with zero decisions and conflicts. That repair postdates the frozen binary and does not change this campaign's negative verdict.

## Next check

Use the repaired source first on the six directly solved MDP rows and the three trajectory-sensitive cabp, Break, and timetable rows. Do not rerun the historical comparison or all four gates until this subset is recovered. The parallel model-guidance work keeps Walk and CDCL as separate actions over the same exact candidate and records availability, installation, use, overwrite, improvement, and export without assigning a reward.

The technical report is benchmarks/reports/ 2026-10-04-restored-lymphosat-sat-regression-terminal.md. Raw evidence remains under the four benchmarks/runs/20261004-restored-lymphosat-sat-* directories.